Algo

① supervised ML → Wrap up — ④

Agenda ⇒ ML ⇒ Algos ⇒

RAG ⇒ → 20% ⇒ 80% ⇒

② → Missing values
→ Handling outliers
→ Encoding
→ Norm | Std.
→ Handle high variance

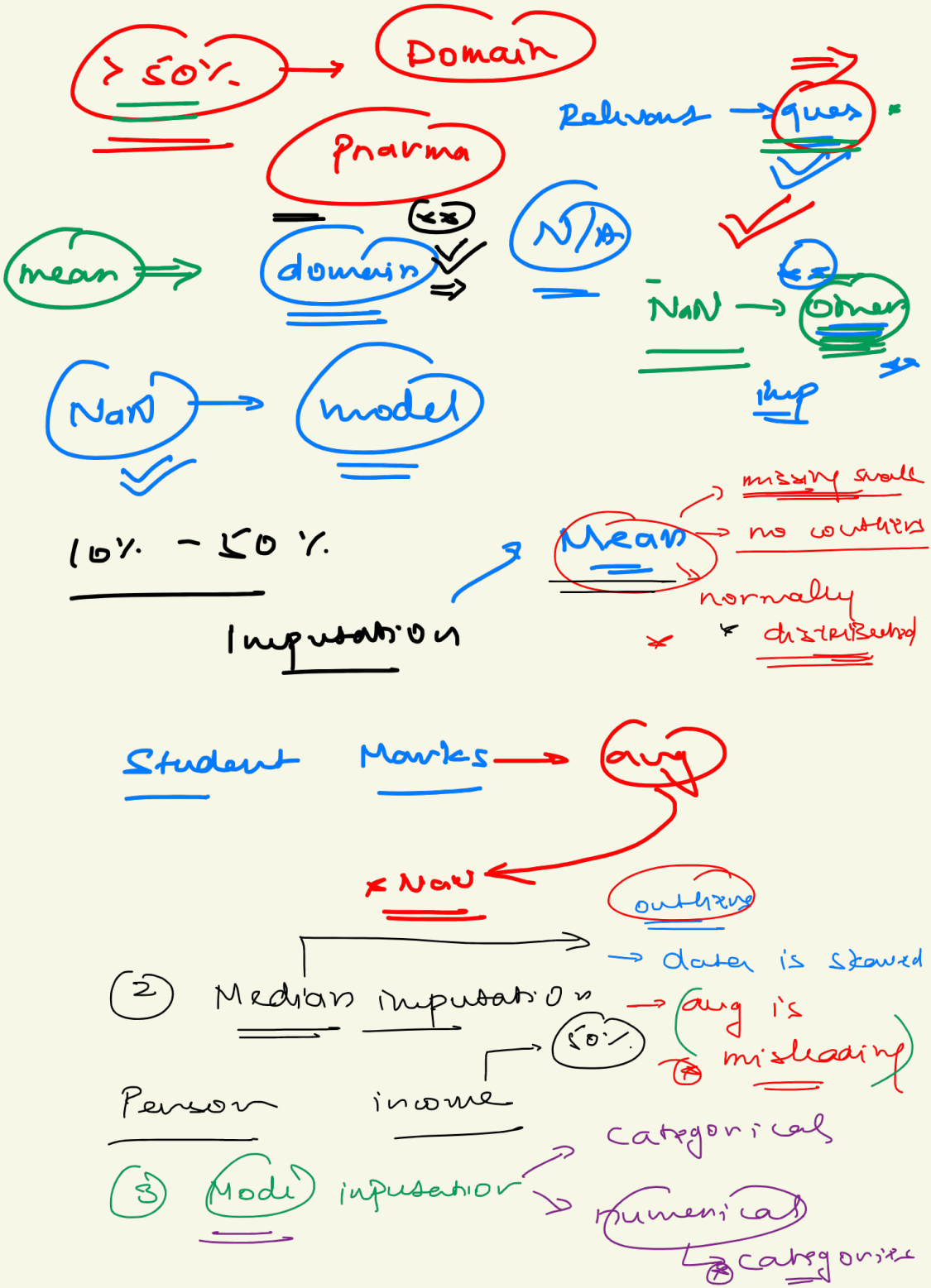
Ques 1 → How do you deal with Missing values in DATA ⇒

1 min. → • > 50% → Drop column ⇒

10000 • upto 10% → Drop These Rows

• 10% – 50% → imputation ⇒

① mean, median, KNN


> 50%
Domain
Pharma
Relevant
ques
mean
domain
N/A
NaN
Other
imp
NaN
model
10% - 50%
Imputation
Mean
missing small
no outliers
normally
distributed
Student
Marks
avg
NaN
outliers
(2)
Median imputation
data is skewed
avg is
misleading
Person
income
50%
(3)
Mode
imputation
categorical
numerical
categorical

[ ] corr()

age year → car price

Rating ⇒ avg.

model

Regression imputation

Domain

Person	Experience	Age	Salary
A	2	24	30000
B	5	28	60000
C	8	32	90000
D	10	35	Missing

⇒ CORRELATION

ML → weights

110000

House	Area	Bedrooms	Location	Price
A	1000	2	Pune	60 lakh
B	1500	3	Pune	90 lakh
C	2000	4	Mumbai	2 crore
D	1800	3	Mumbai	Missing

does not impact prediction

Null

Lin Reg ⇒

Scatter plots

Regression ⇒ lots

CORRELATION

KNN imputation ⇒

k=2 ⇒

categorical

Student	Study Hours	Attendance	Marks
A	2	60%	40
B	3	65%	45
C	8	90%	85
D	9	92%	88
E	8	91%	Missing

avg

86 / 87

Customer	Age	Monthly Spending	Website Visits	Membership Type
A	23	2000	10	Basic
B	25	2200	12	Basic
C	45	9000	30	Premium
D	47	8500	28	Premium
E	46	8800	29	Missing

K=2

KNN

KNN

Premium

Problem → Expensive

→ Too much missing → Drop

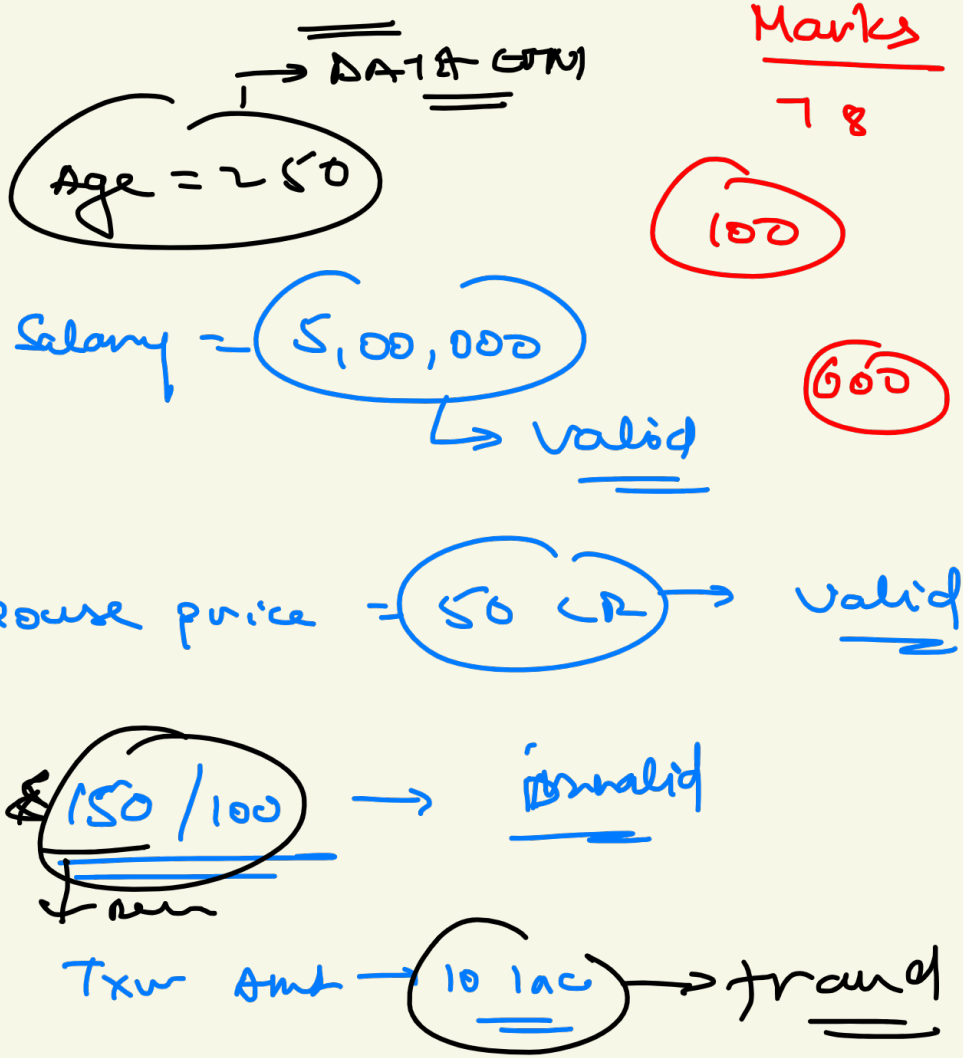
→ very little missing data → drop the rows

Moderate Missing → Imputation

Imputation

Situation	Best method
Numerical, normal distribution, no outliers	Mean imputation
Numerical, skewed distribution, outliers present	Median imputation
Categorical data	Mode imputation
Missing value related to other variables	Regression imputation
Similar records should have similar values	kNN imputation

Ques 2) * Diff. ways to deal with outlier values?



Removal → (1) Remove

(2) Z Score

(*) Dist. from mean / σ

<<

0 → avg

1 → 1σ away

2 → 2σ

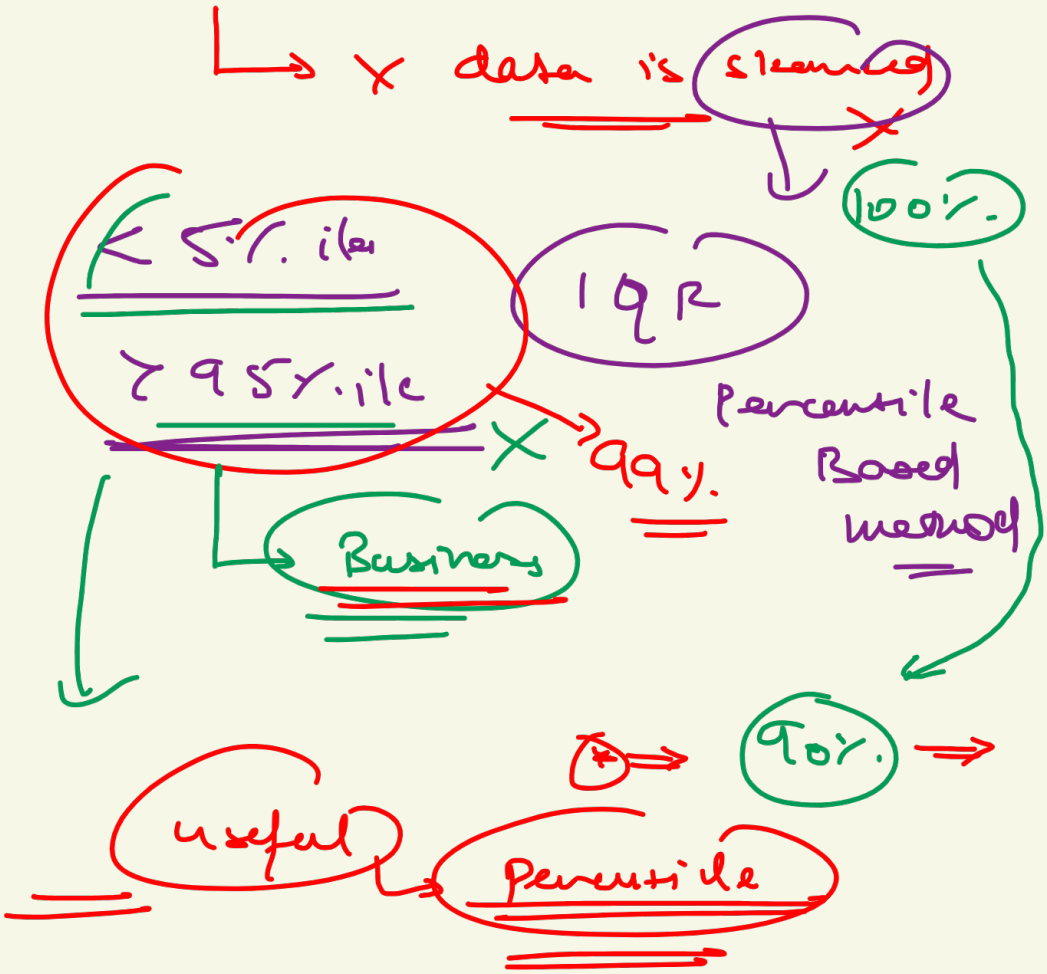
3 → very far

> → Remove

⇒ upper limit = mean + 3σ

lower limit = mean − 3σ

useful → Normally dist (**)


X data is skewed
100%.
< 5%.ile
> 95%.ile
IQR
99%.
Percentile Based method
Business
90%.
useful
Percentile

1 — 99

5 — 95

10 — 90

IQR → method

Q_1 → 25th %ile

Q_2 → median Q_3

IQR → $Q_3 - Q_1$ (sort)

lower = $Q_1 - 1.5 \times IQR$

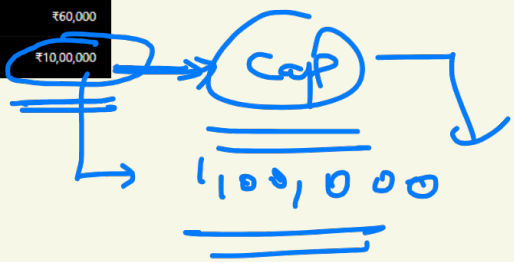
upper = $Q_3 + 1.5 \times IQR$

→ Skewed ⇒ Tree → ✗✗

→ Capping ⇒ less →

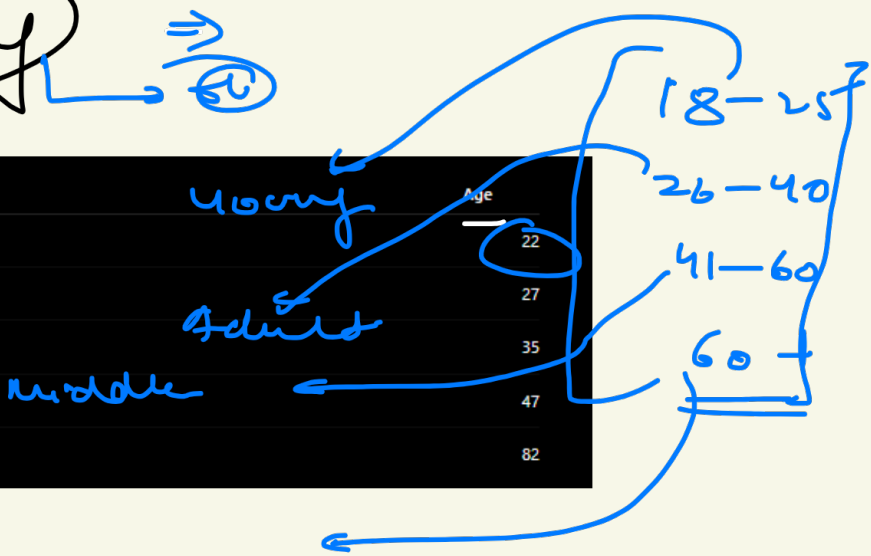
→ clipping or Flooring

Customer	Income
A	₹30,000
B	₹40,000
C	₹45,000
D	₹50,000
E	₹60,000
F	₹10,00,000





Person	Age
A	22
B	27
C	35
D	47
E	82





transformation

(1) Log

(2) Reciprocal → 1/value

(3) Sqrt

(4) Box Cox

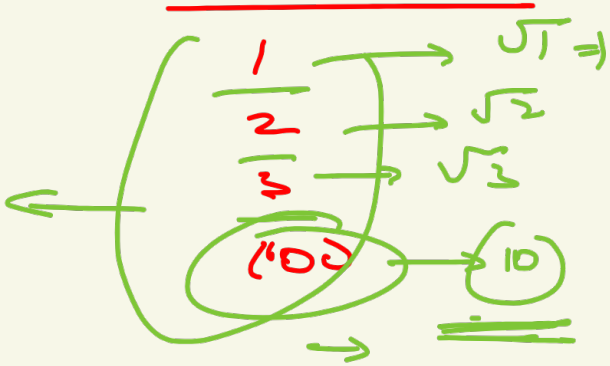
log(

1/2	1
10 →	0.1
100	0.01
1000	0.001

Customer support Tickets

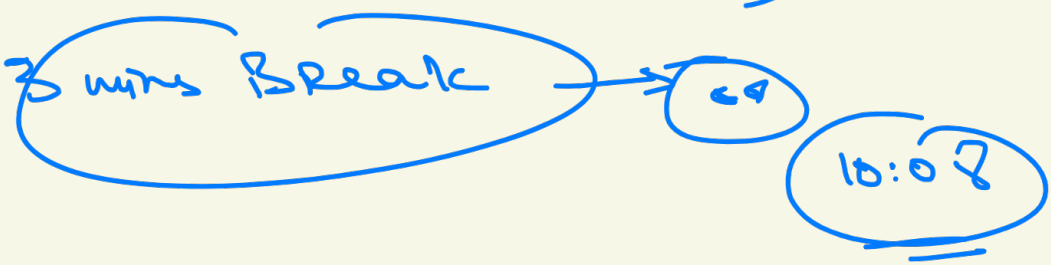
Case

Tickets Raised



→ missing, outliers

Method	Intuition
Removal	"This value is wrong or not useful. Delete it."
Z-score	"This value is too far from the average."
Percentile	"This value is among the most extreme top/bottom values."
IQR	"This value is far outside the normal middle range."
Capping/Flooring	"Keep the row, but limit the extreme value."
Transformation	"Compress the scale so extremes hurt less."
Binning	"Exact value does not matter; only the range/category matters."



→ Normalization Vs Std.

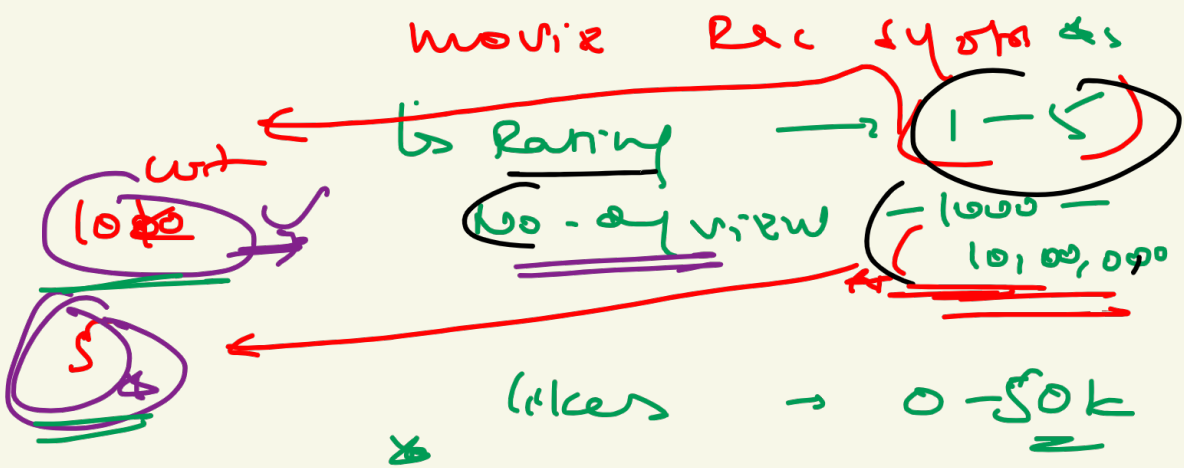
↳ when?

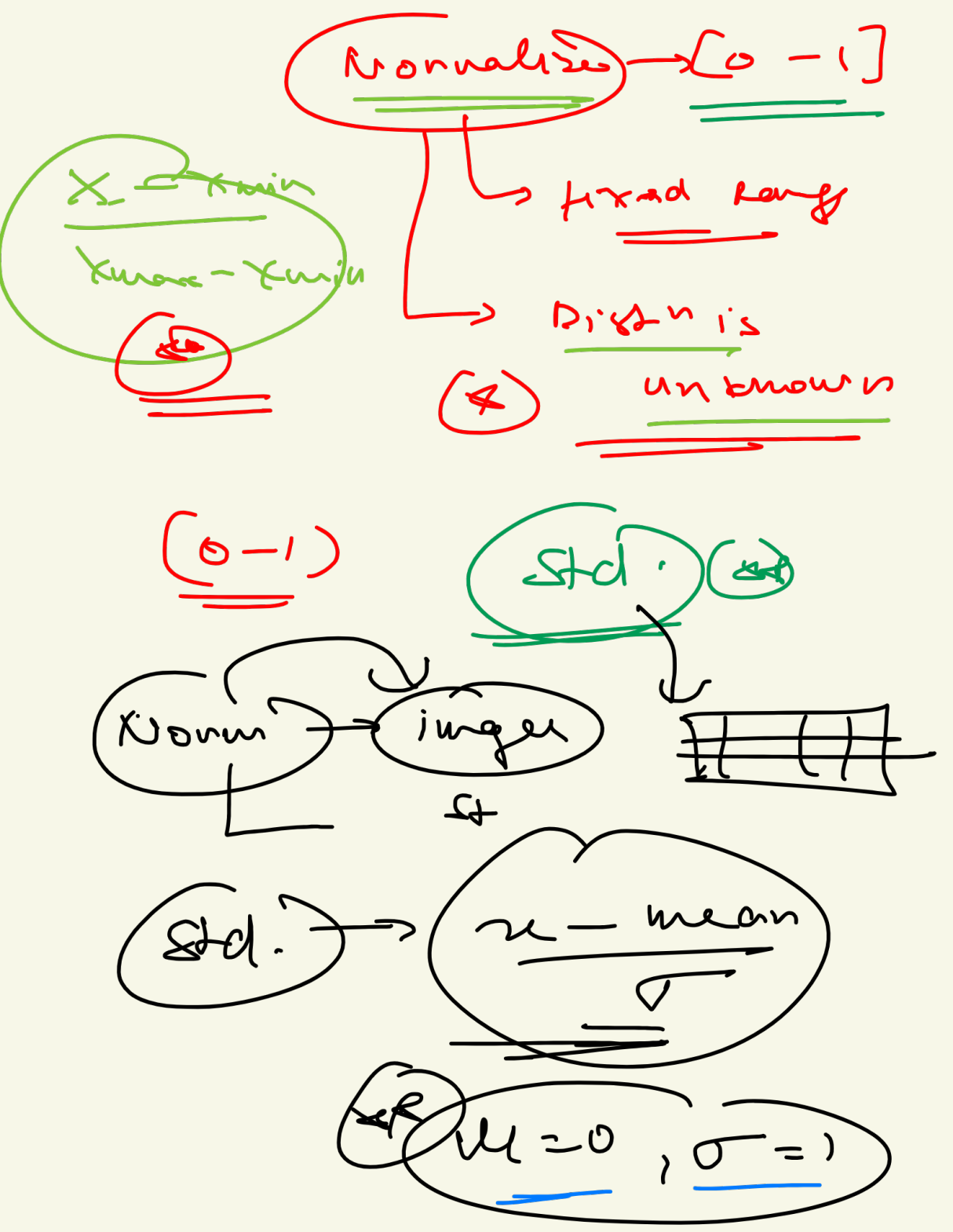
& (min →

Normalization

↳ Min max scaling

↳ [0 - 1]




Normalize → [0 - 1]
fixed Range
Digit is
unknown
[0 - 1]
Std.
Norm
image
Std.
μ — mean
σ
μ = 0 , σ = 1

Std. → data is approx. Gaussian

→ ~

→ X

img → 0–255

→ 0

dist. → x

Std.

lin reg
log reg
SVM
PCA

GD

is Std. sensitive to outliers

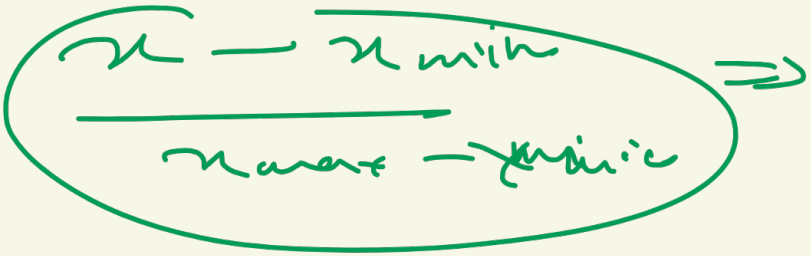
less sensitive

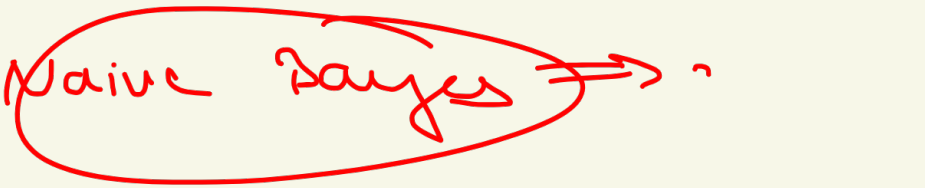
Min max

Normalization

20, 40, 60, 80, 1000

mean ⇒









(0 — 1)





Ques → When should we use which type of encoding

- → Does cat have order?
- → How many unique cat.?
- → Does freq. matter
- → Does cat. have Rel.^ with Target
- → Is it cat. or textual data

Message

Text



Step 1 → full sentence

→ Embedding → ④

Step 2 → Does category have order

low < med < high

↓ ↓ ↓

1 2 3 → Ordinal

→ ② Label Encoding

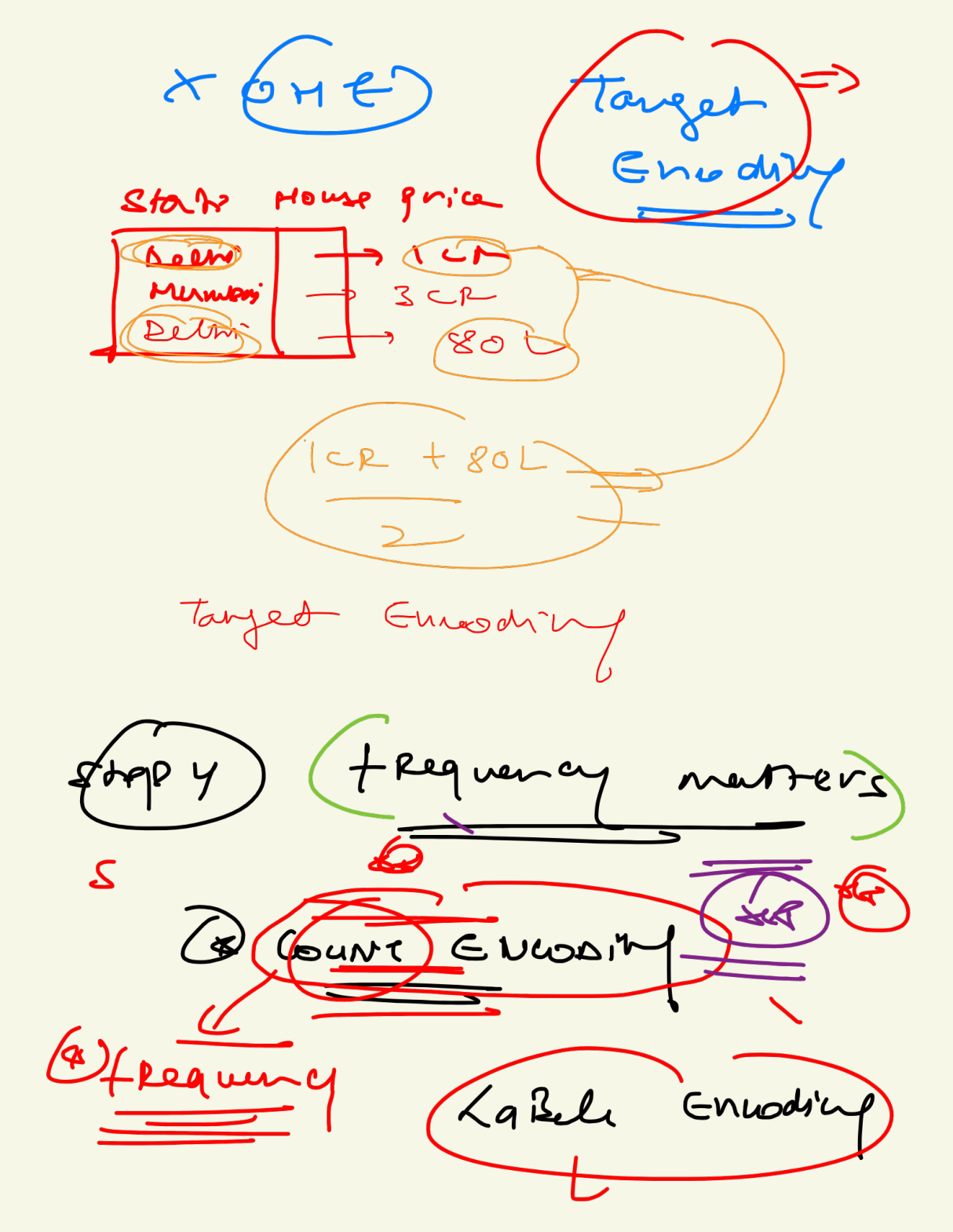
Step 3 → unique categories

⑤ Payment method

↙

UPI, Debit/Credit/Cash

→ OHE


Target Encoding
Start House price
Delhi
Mumbai
Delhi
1 CR
3 CR
80 L
1 CR + 80 L
2
Target Encoding
Frequency matters
Count Encoding
Frequency
Label Encoding

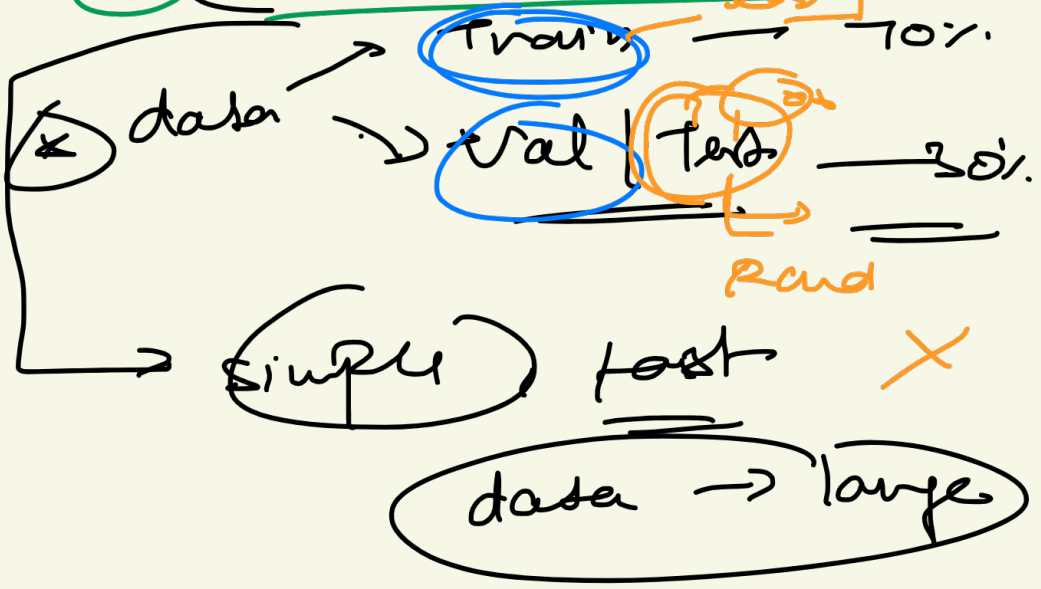
Steps → Relv cat v/s Target

→ Targets Encoding

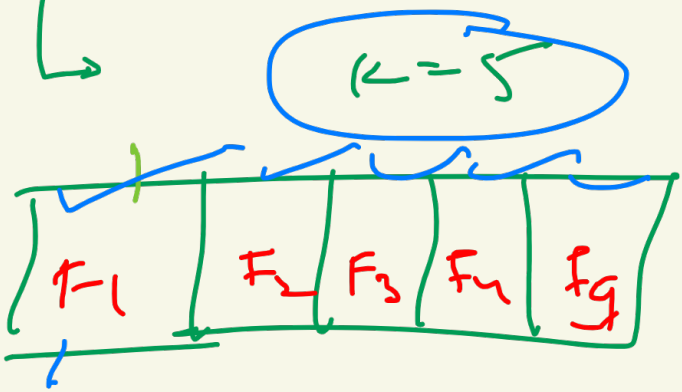
Ques → What are various CV techniques we can use

min →

(1) Holdout validation



(2) K fold CV



Round 1 →

Vali → F1

Train → F2 + F3 + F4 + F5

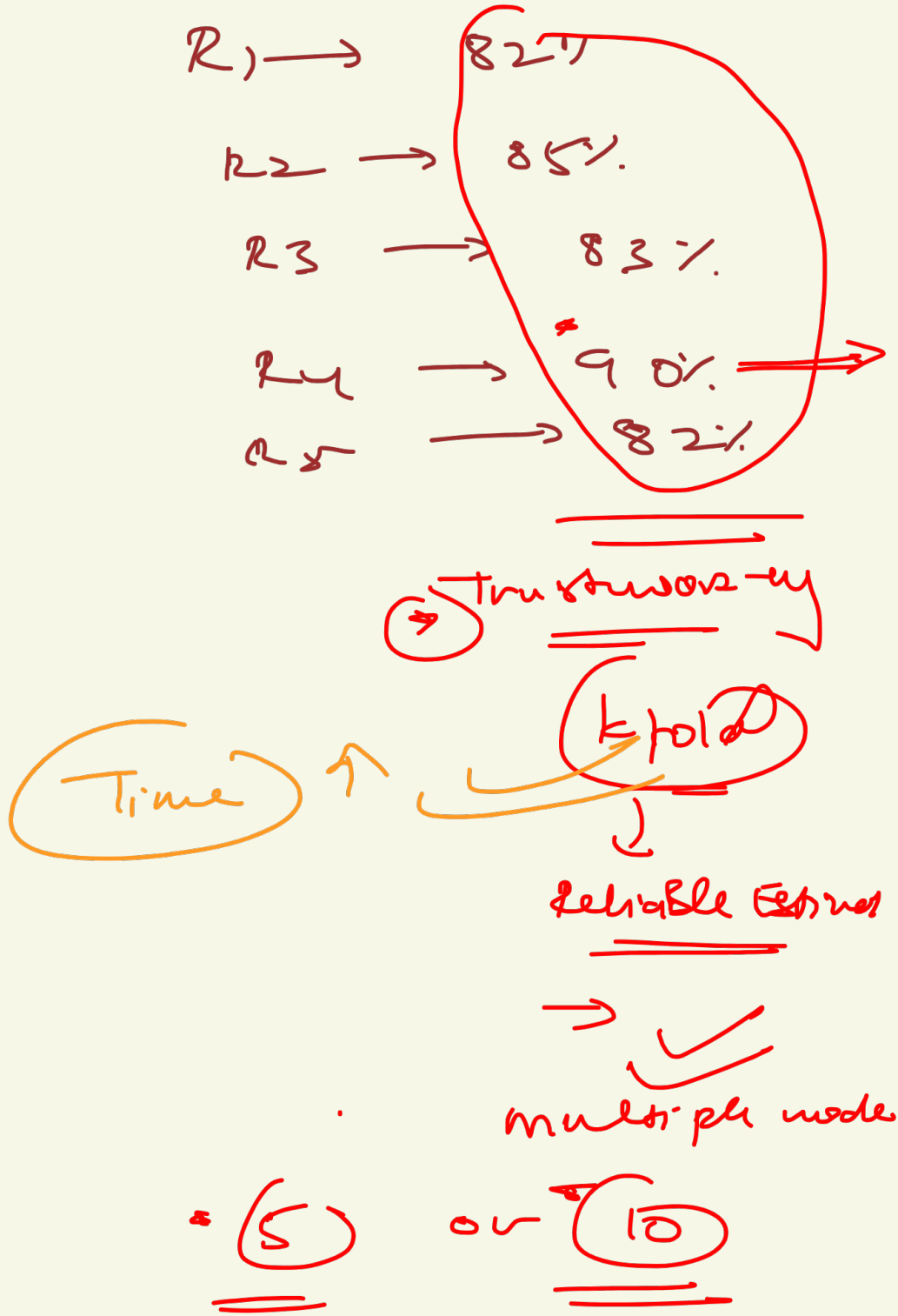
Train

(2) →

Val → F2

Train → F1 F3 F4 F5

(5) th Round


R1 → 82%
R2 → 85%.
R3 → 83%.
R4 → 90%.
R5 → 82%.
Transmission
threshold
Time
Reliable Efficient
multiple node
5 or 10

Stratified k fold →

imb data

NF → 900

F → 10

100

k fold

R1

90 NF
10 F

Train → 18 NF

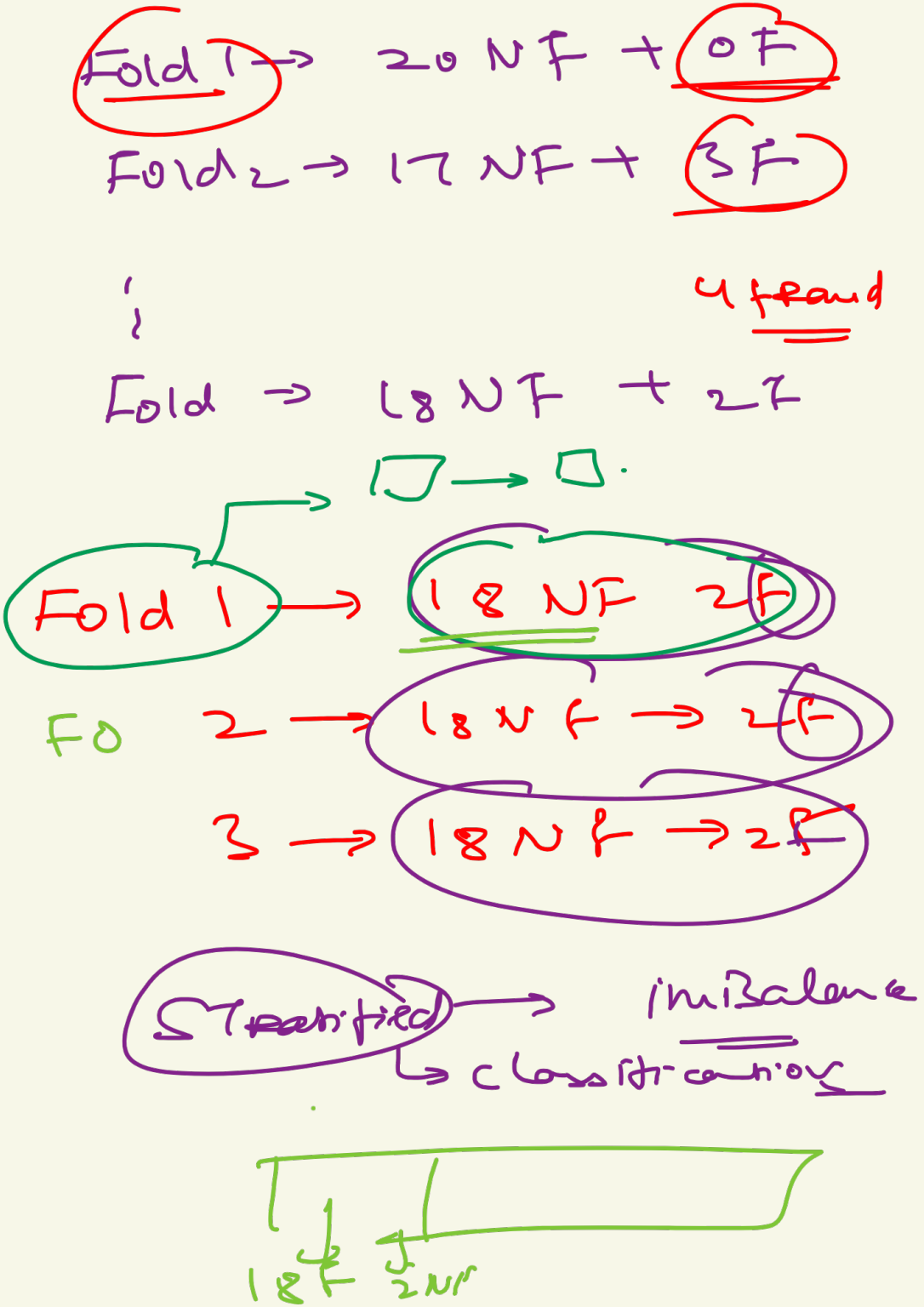
val → 2 F

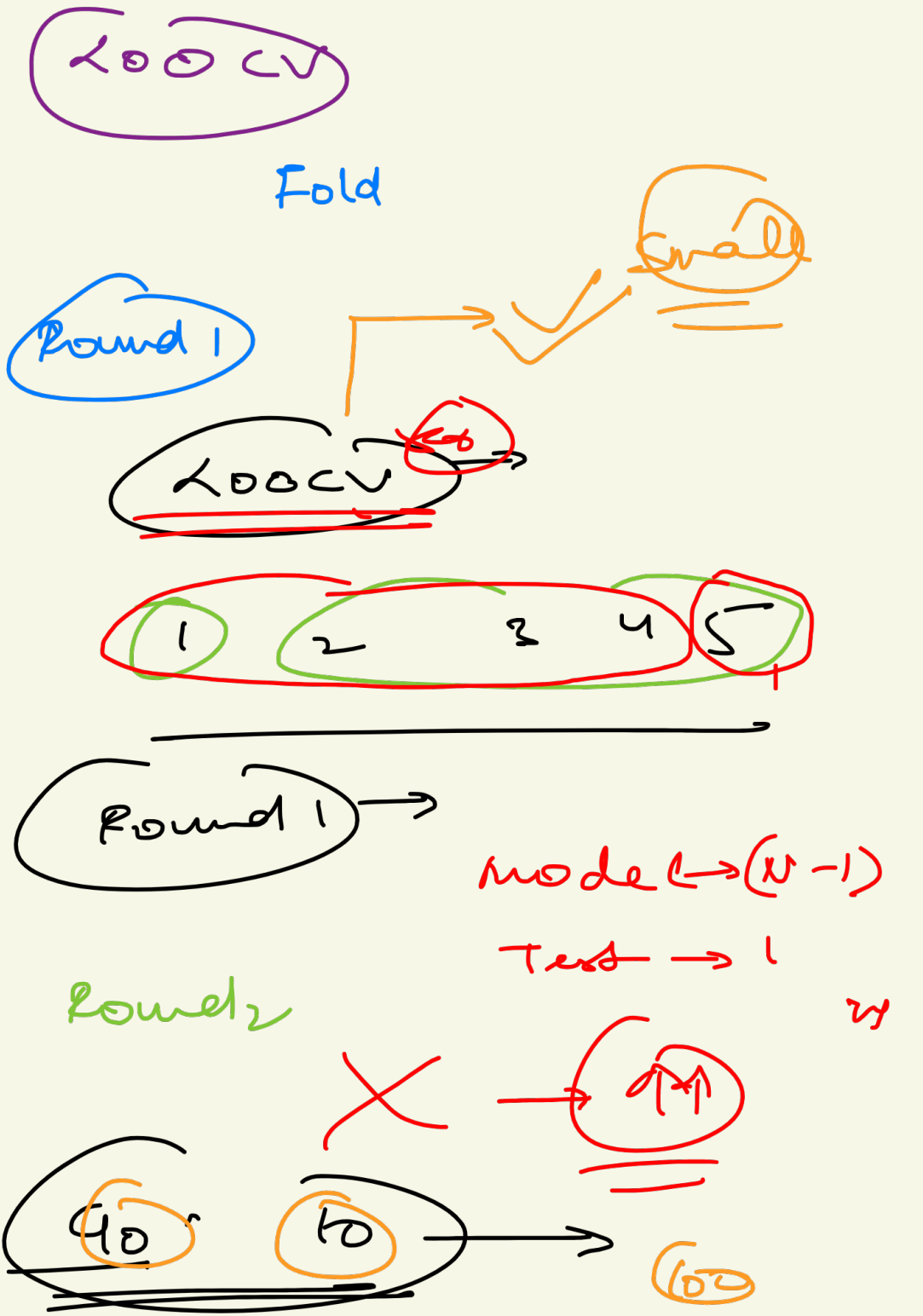
Overfit

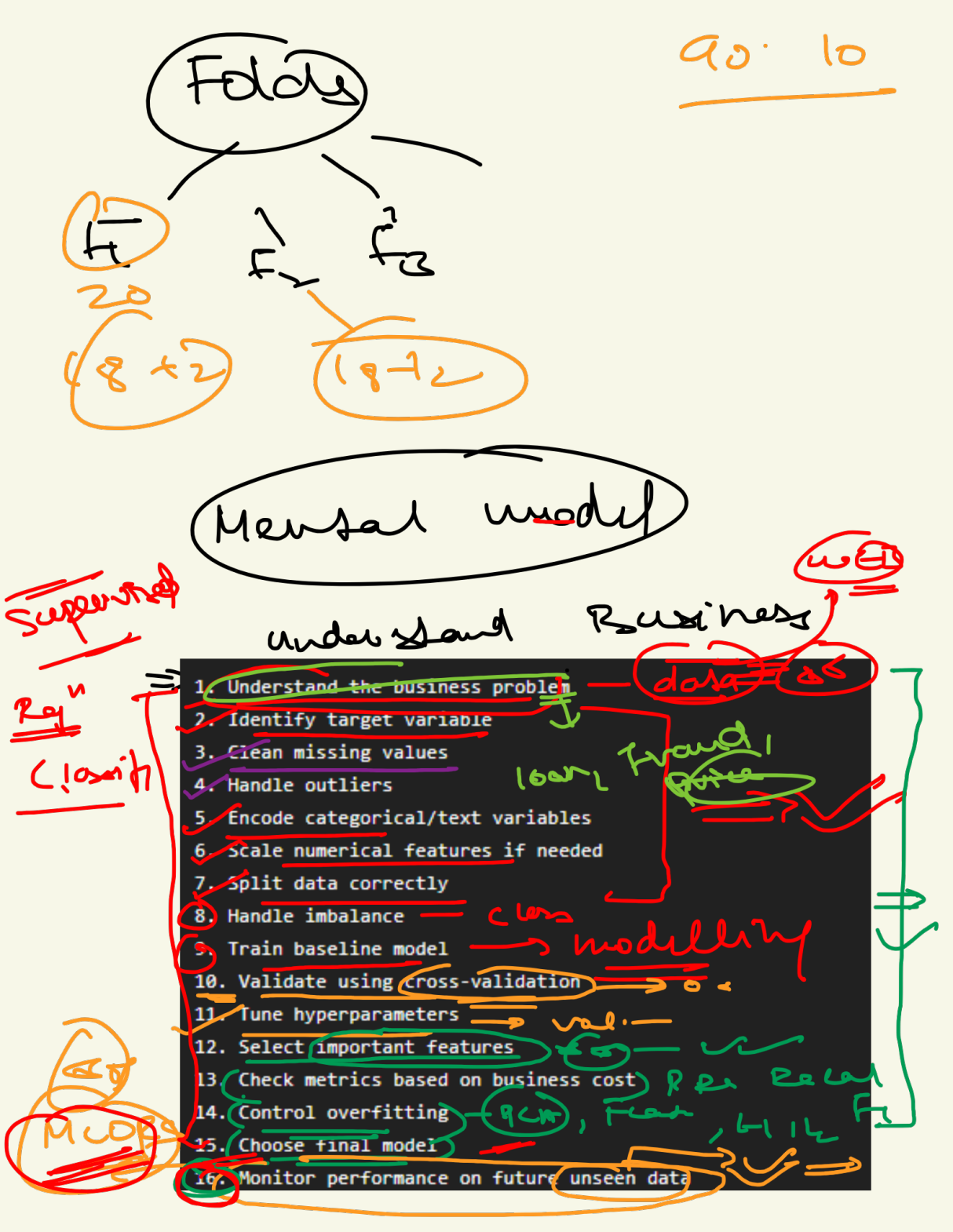
imb

validation

F1 →


Fold 1 → 20 NF + 0F
Fold 2 → 17 NF + 3F
Fold → 18 NF + 2F
4 fraud
Fold 1 → 1 & NF 2F
Fo 2 → 1 & NF → 2F
3 → 1 & NF → 2F
Stratified → imBalance
classification
1 & F
2 NF


LOOCV
Fold
Round 1
Overall
LOOCV
1
2
3
4
5
Round 1
Model → (N-1)
Test → 1
Round 2
40
40
100


Folds
Mental model
Supervised
understand
Business
1. Understand the business problem
2. Identify target variable
3. Clean missing values
4. Handle outliers
5. Encode categorical/text variables
6. Scale numerical features if needed
7. Split data correctly
8. Handle imbalance
9. Train baseline model
10. Validate using cross-validation
11. Tune hyperparameters
12. Select important features
13. Check metrics based on business cost
14. Control overfitting
15. Choose final model
16. Monitor performance on future unseen data